PRODUCT PROPOSAL / 2026

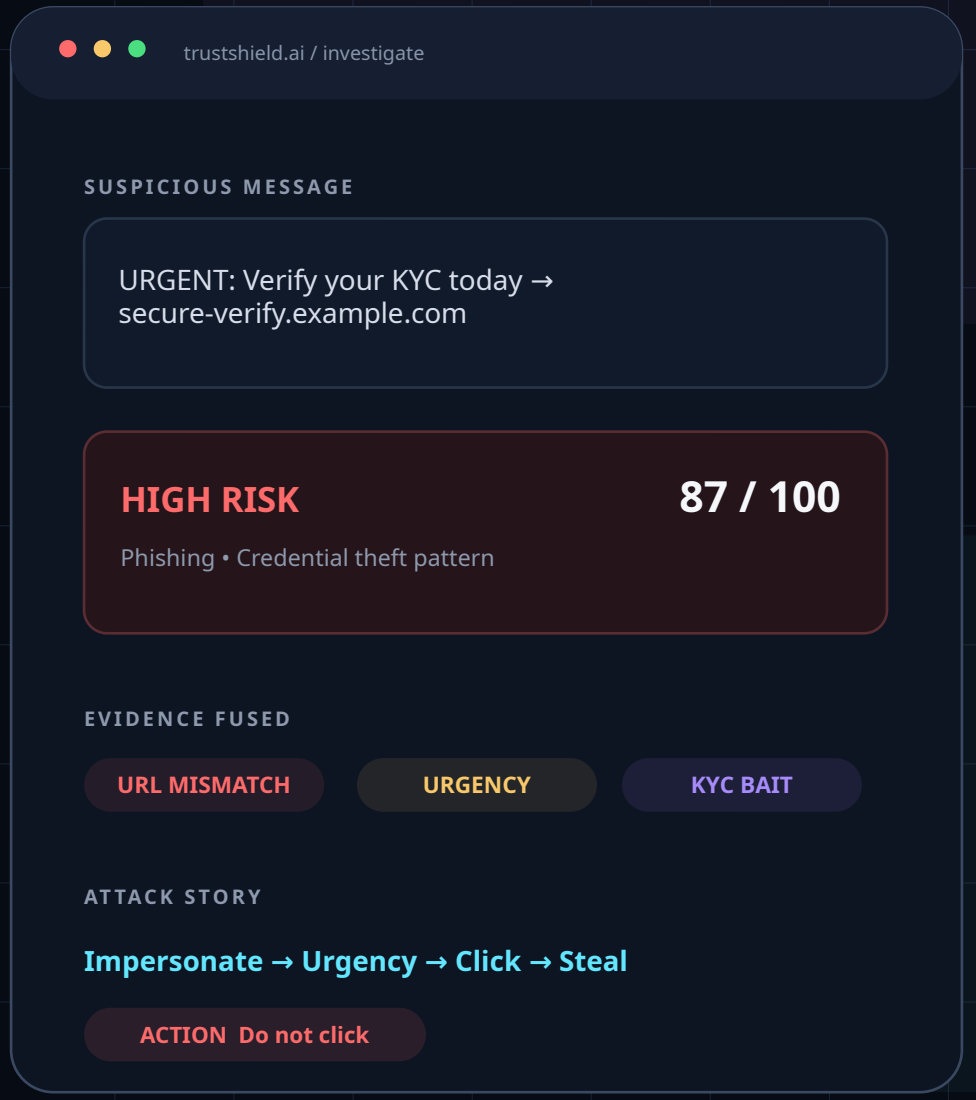
TRUSTSHIELD AI

The cybersecurity agent that does more than say “phishing.”

Investigate the message. Reconstruct the attack.
Recommend the safest next action.

AI / ML · CYBERSECURITY · AI AGENT

TEAM Aryan Mishra · Ayush Bajpai · Ayush Gupta



The verdict is not the problem. The uncertainty is.

People encounter suspicious digital messages every day — but a “phishing” label alone does not tell them why, how, or what to do next.

REAL-WORLD SCENARIO

**“Your account will be blocked today.
Verify KYC now.”**

secure-verify.example.com

A user immediately needs 4 answers:

01 Is this safe?

02 What looks suspicious?

03 How is the attacker manipulating me?

04 What should I do next?

425,808 phishing attacks tallied by APWG in June 2026 — the highest monthly total since Apr 2023.

+40% smishing growth from Q1 → Q2 2026

Source: APWG, Phishing Activity Trends — Q2 2026

FRAGMENTED SIGNALS

URL reputation, message cues and security databases live in different places.

LOW EXPLAINABILITY

A binary verdict rarely helps a non-technical user understand the attack.

NO ACTION LOOP

The user is left to decide whether to click, pay, reply or report.

TrustShield is a decision engine, not another phishing detector.

One secure workspace turns a suspicious message or URL into evidence, an attack story, and a clear next action.

INPUT

SMS | EMAIL | URL | SCREENSHOT

01 INSPECT

Extract the signals

URLs • domains • urgency • impersonation • requests • risky cues

02 EXPLAIN

Reconstruct the attack

Fuse technical evidence with social-engineering context and produce a human-readable attack story.

03 PROTECT

Recommend the next step

Tell the user what to avoid, verify, preserve or report — before they act.

CORE PRODUCT PROMISE **“Don’t just detect the scam — show the user how the attack works and what to do next.”**

One suspicious message becomes a defensible security decision.

The demo is designed to be understood in one glance: every conclusion is backed by visible evidence.

STEP 1 / INPUT

Suspicious message

URGENT: KYC expires today.
Verify now →

secure-verify.example.com

The user can paste text, a URL, or upload a screenshot.

SCAN

EXTRACT
Text + URL signals

ENRICH
Reputation + heuristics

FUSE
Weighted evidence

EXPLAIN
Attack story

TRUSTSHIELD REPORT

HIGH RISK

87 / 100

Why this is risky

- **Domain mismatch** — The destination does not match the claimed brand.
- **Urgency cue** — The message pressures the user to act immediately.
- **Credential bait** — The flow is designed to collect verification data.

SAFE ACTION

Do not click

From a verdict to an attack story.

Our differentiator is the investigation layer that turns existing security signals into an understandable, actionable decision.

TRADITIONAL FLOW

URL → reputation check → verdict

- ✓ Detects known bad URLs
- ✓ Aggregates security signals
- → Often leaves interpretation to the user
- → Does not close the decision loop

TRUSTSHIELD FLOW

Message + URL → evidence → story → action

01	EVIDENCE FUSION	Multiple visible signals support the score.
02	ATTACK STORY	Explain how the manipulation is intended to work.
03	ACTION LOOP	Translate the verdict into a safe next action.

SIGNATURE DIFFERENTIATOR

ATTACK STORY RECONSTRUCTION

Evidence → story → action

TEAM TRACK RECORD

EcoClean AI

Smart Waste Management System • Code X2

smart-waste-management-sandy.vercel.app

Reference examples: Google Safe Browsing • VirusTotal URL intelligence

One repo. One backend. Thin integrations.

Optimized for a 12-hour build: single codebase, thin integrations, no unnecessary microservices or model-training pipeline

USER EXPERIENCE

- **Next.js / React**
- Tailwind CSS
- shadcn/ui-style components

APP LOGIC

- **Next.js server routes**
- TypeScript
- Zod validation

INTELLIGENCE

- **Groq-hosted LLM**
- Agent prompt + tools
- Evidence scoring

SECURITY SIGNALS

- **URL heuristics**
- VirusTotal adapter
- Safe Browsing adapter

DATA / DEPLOY

- **Supabase Postgres**
- Vercel
- GitHub

12-HOUR DESIGN RULE

Use the LLM for explanation • deterministic rules for safety-critical checks • external APIs only as enrichments • graceful fallback when APIs are unavailable.

Docs: Groq API • Supabase + Next.js • VirusTotal URL API • Google Safe Browsing

We can ship the demo — because the scope is designed around the clock.

The prototype focuses on one polished user journey instead of a long feature list.

12-HOUR EXECUTION MAP

00–02 FOUNDATION

Scaffold UI, routes, schema, env.

02–05 CORE SCAN

Input → parse → URL heuristics → report shell.

05–08 INTELLIGENCE

Groq agent + evidence fusion + attack story.

08–10 POLISH

Demo states, charts, copy, responsive UI.

10–12 QA / DEMO

Run tests, verify fallbacks, rehearse pitch.

MVP — ONE GOLDEN PATH

1. Paste message / URL
2. Extract risk signals
3. Run AI investigation
4. **Show evidence + risk score**
5. **Generate attack story**
6. Recommend safe action

FEASIBILITY GATES

NO MODEL TRAINING	LLM + rules + reputation signals
NO MICROSERVICES	Single Next.js application
NO HARDWARE	Software-only prototype
DEMO-FIRST	One complete user journey

Built to communicate the product in seconds.

TrustShield combines AI investigation with practical cybersecurity protection in one user journey.

AI / ML

LLM-based investigation + evidence classification

AI AGENT

Tool-using workflow that gathers signals and explains them

CYBERSECURITY

Phishing • fraud • privacy • threat detection

SUCCESS IS NOT "MODEL ACCURACY" ALONE.

We measure whether a user can move from uncertainty → informed action.

1	**Evidence clarity**	Can the user see why the score was given?
2	**Action clarity**	Can the user identify the safest next step?
3	**Demo reliability**	Does every test input return a structured report?

THE TAKEAWAY

Detect.

Explain.

Protect.

TRUSTSHIELD AI

BUILD TEAM **Aryan Mishra · Ayush Bajpai · Ayush Gupta**

Software-first • AI-assisted build • 12-hour MVP discipline